

The EV Charging Intelligence Framework

A Decision System for Capital and Operational Deployment in Indian Tier-1 Cities

Bengaluru Case Study — April 2026

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1. Executive Summary

The opportunity

India's public EV charging network has grown from roughly 5,000 stations in FY21 to over 29,000 stations by end-FY25 — a sixfold expansion in four years (IMARC, 2026). Against that, NITI Aayog and industry consensus estimate a 2030 requirement of approximately **3.9 million public charging stations**. The country is running at less than 1% of target coverage. Even under optimistic assumptions about private-sector acceleration, this implies an installation pipeline of roughly 750,000 stations per year for the next five years — a capital and operational challenge of several lakh crore rupees in aggregate.

Government has committed ₹10,900 crore through PM E-DRIVE. Private capital is arriving. The technology is mature. The remaining question is allocation: **given finite capital and limited operator bandwidth, where should the next charger go?**

The problem with current practice

As of 2026, Indian CPOs (Charge Point Operators) deploy charging infrastructure through what IMARC Engineering, a consulting firm serving Indian infrastructure developers, describes as a supply-driven process — land availability first, demand verification second or not at all (IMARC, 2026). The predictable consequences are visible across every Tier-1 city:

- **Overcrowded hotspots** at tech parks and malls, with peak-hour wait times of 25–45 minutes
- **Stranded capacity** in residential zones, with reported utilisation of 8–12%
- **Charging deserts** in rapidly growing peri-urban corridors where EV adoption has outpaced infrastructure

Each misplaced fast charger represents ₹15–25 lakh of unrecoverable capital. IMARC's own field observation — that "site selection determines 60–70% of a charging station's long-term utilisation rate" (IMARC, 2026) — is the industry's own admission that this is the dominant decision variable, and simultaneously the decision most operators make worst.

What this report contributes

This report presents a decision framework for EV charging deployment that addresses two distinct but coupled decisions:

The CAPEX decision — where to build permanent stations. A demand-weighted, accessibility-corrected Congestion Index (CI) ranks micro-zones within a city by their unmet demand pressure. A composite Site Score layers in commercial accessibility, grid stability, and forward-looking growth to produce an executable deployment ranking.

The OPEX decision — where to deploy mobile CaaS units next week. A CaaS Deployment Score (CDS) captures transient visitor demand, time-resolved footfall, and the short-term competitive landscape. CDS operates weekly rather than annually, and optimises for mobile unit revenue capture — a problem structurally different from permanent siting.

The coupling between the two — the central thesis — is that **mobile CaaS deployment is the demand-validation instrument that de-risks CAPEX commitment**. A 30-day mobile pilot costs a few thousand rupees, produces empirical utilisation data, and converts a ₹12–13 lakh CAPEX commitment under uncertainty into one informed by real observations. CaaS is not a substitute for CAPEX; it is the feasibility test that precedes CAPEX. No existing consulting framework or academic paper addresses this coupling at the operator-decision level — IMARC's 20-section comprehensive guide to Indian EV charging infrastructure does not cover CaaS as a distinct product category or decision instrument (IMARC, 2026). This is a genuine white space.

What is demonstrated

A seven-module Python data pipeline collects publicly available data on EV registrations, charger supply, points of interest, footfall proxies, land prices, and real-world deployment patterns. The pipeline computes CI and CDS at ~2 km micro-zone granularity for Bengaluru. A Random Forest Regressor forecasts congestion 12 months ahead. A financial engine translates scores into rupee payback periods under multiple utilisation scenarios.

The demonstration runs on a calibrated synthetic 20-zone dataset for Bengaluru, validated partially against a real scraping of 56 Bengaluru charging stations from eHUB by MG — the aggregator platform that integrates 13 CPOs

and covers 2,000+ stations nationally.

What the numbers say

Under the framework's CAPEX scoring:

- HSR Layout Sector 1 and Outer Ring Road emerge as top-ranked CAPEX zones despite having zero DC fast chargers in the real scraped data — the classic charging desert pattern
- Marathahalli, with 6 DC fast chargers from Shell and other operators, correctly drops to lowest-priority (saturated zone)
- Peri-urban zones (Hoskote, Devanahalli, Yelahanka) rank middle-tier — growth potential offset by current low demand

Under the framework's financial model, calibrated to IMARC's 2026 cost breakdown and PM E-DRIVE subsidy rates:

- Net CAPEX after subsidy for a 2-charger 50kW station: **₹12.6 lakh**
- At realistic Indian urban utilisation (25% / ~6 sessions per charger per day): **break-even in 4-5 years** (IMARC, 2026), or 18-20 months on net CAPEX after subsidy
- At premium-site utilisation (40% / ~10 sessions per charger per day): **break-even in 7-10 months** on net CAPEX

What the operator should do

The report recommends a three-stage deployment sequence for Bengaluru CPOs:

1. **Screen** candidate zones using the CI-ranked list (top-15 by Site Score)
2. **Validate** top-ranked zones with 30-day CaaS pilot deployments — CDS directs the pilot
3. **Commit** permanent CAPEX only where observed utilisation supports payback below operator target threshold (typically 18 months net-of-subsidy)

This sequence reduces the stranded-asset rate materially. A single mis-sited permanent station wastes ₹12-13 lakh; a failed 30-day CaaS pilot wastes a few thousand rupees of transport and setup. The information value of CaaS as a probe, relative to its cost, is extraordinary.

What is not claimed

The current demonstration uses calibrated synthetic data, partially validated against real eHUB scrapes. Specific zone rankings are illustrative of the framework's mechanical behaviour, not field-validated deployment recommendations. Converting the framework into an operational tool requires the data upgrades and validation steps specified in Section 7 — principally, an operator partnership to access CSMS utilisation data and a real 30-day mobile pilot in a top-ranked zone.

2. Market Context — Why Now

2.1 The scale gap

India’s EV market has crossed a threshold. In FY25, EV adoption reached the following penetration rates across vehicle segments (IMARC, 2026):

- **Two-wheelers:** 6.4% of total sales
- **Three-wheelers (passenger):** 31.8%
- **Three-wheelers (cargo):** 21.6%
- **Four-wheelers:** 4.3%

The four-wheeler number is the one that matters for public DC fast charging. It is small today but growing steeply — new model launches from Tata, MG, BYD, Hyundai, and Mahindra have begun closing the model-availability gap that constrained 2020–23 demand. Tata Motors’ passenger EV range alone now covers price points from ₹8.5 lakh to ₹22 lakh on-road.

Meanwhile, public charging infrastructure is progressing but not at the required pace:

Metric	FY21	FY25	2030 Target (IMARC/industry)	2030 Target (NITI Aayog)
Public charging stations	~5,000	29,000+	3,900,000	444,000
Ratio of EVs to public stations	1:88	1:235	1:12 (target)	—
Tracked growth	—	5.8× in 4 years	134× from FY25	15× from FY25

The discrepancy between NITI Aayog’s earlier 4.44 lakh figure and the industry-consensus 3.9 million figure reflects different assumptions about 2030 four-wheeler penetration and Tier-2/3 city coverage. This report adopts the 3.9 million figure as the planning baseline because it matches how CPOs and infrastructure investors are framing the opportunity in 2026 private-market discussions.

At either target, the deficit is severe. India needs to deploy, at minimum, **roughly 95,000 new public charging stations per year** between 2025 and 2030 to reach the NITI Aayog target, and over **750,000 per year** to reach the industry consensus target. The current run-rate is approximately 30,000–40,000 per year. Acceleration of 2–25× is required. This is the single largest infrastructure allocation decision in Indian transportation over the next five years.

2.2 The commercial signal

Public capital cannot fill this gap alone. The government’s ₹10,900 crore PM E-DRIVE commitment — of which ₹2,000 crore is allocated specifically to 72,300 charging stations (22,100 DC fast for four-wheelers, 1,800 for buses, 48,400 for two- and three-wheelers) — finances less than 2% of the 3.9 million target (IMARC, 2026). Private CPO capital must provide the remainder.

The private capital is arriving. Recent movements (2025–2026):

- **Maruti Suzuki** launched a unified digital charging platform (December 2025) integrating 13 CPOs and 2,000+ stations across 1,100 cities, targeting 100,000 stations by 2030 (IMARC, 2026)
- **HPCL × V-GREEN** announced (December 2025) a rollout of EV charging across HPCL’s 24,400+ fuel retail outlets — by far the largest single distribution partnership in the sector
- **MobiLane** announced a target of crossing 1,000 stations by end-2026, from ~250 installed
- **Incharz × NPCL** opened multi-vehicle DC fast charging plazas in the Delhi NCR belt
- **VNT** introduced India’s first 1 MW EV charging solution (October 2025) for heavy commercial vehicles

The infrastructure-investor case is strong enough that the question is no longer “is this a good market?” but “which sites, in what sequence, at what return?” That is the allocation problem this report addresses.

2.3 Why the allocation problem is hard

Three structural reasons:

First, demand is heterogeneous and mostly invisible. Vahan registrations give you EV counts at the RTO (district) level but not at ward or street level. Visitor demand — the EVs that drive into a zone for work, shopping, or transit — is entirely absent from Vahan. A framework that ranks zones purely on resident registrations is systematically blind to the commercial hubs where DC fast chargers actually earn.

Second, supply is distributed and not uniformly visible. The eHUB by MG aggregator surfaces 2,000+ stations but only about 70 appeared in the Bengaluru scrape this project ran — the true Bengaluru station count including Ather Grid, Statiq, and smaller operators likely exceeds 200. Any framework that uses only one data source systematically miscounts supply.

Third, the two most important variables — zone-level utilisation and operator revenue — are proprietary to each CPO’s CSMS (Charging Station Management System) and not publicly available. This is the central data gap in the Indian market and the reason no operator-ready siting tool exists today.

2.4 What IMARC and adjacent consulting firms cover — and do not

Industry consulting — exemplified by IMARC Engineering’s February 2026 guide, currently the most comprehensive public document on Indian EV charging infrastructure setup — addresses cost, regulation, subsidies, technology selection, grid integration, and operations (IMARC, 2026). It does not address:

- **Zone-level demand prediction at the ~2 km micro-zone scale** — consulting outputs stay at city or state granularity
- **Forward-looking demand modelling with ML and explainability layers** — industry uses trailing demand and operator intuition
- **Mobile CaaS deployment as a distinct product category or decision framework** — the entire 20-section IMARC guide does not reference mobile charging-as-a-service
- **The coupling between mobile pilot deployment and permanent CAPEX commitment** — treated as entirely separate business activities

These four are the contributions of the present framework. The CaaS gap in particular is a **genuine white space** — the Indian market has CaaS product offerings (Teslavlots, NITA Energy, various regional players) but no shared analytical framework for where to deploy mobile units or how mobile pilots should inform subsequent permanent investment.

2.5 Who this framework serves

The framework is designed for three decision-makers, with overlapping but distinct use cases:

Audience	Primary use
CPO operations teams	Weekly CaaS routing (CDS output); quarterly CAPEX deployment ranking (Site Score output)
CPO investment committees	CAPEX portfolio allocation across top-ranked zones; sensitivity-aware payback analysis
State transport and urban planning authorities	PM E-DRIVE subsidy targeting; zoning mandates for EV-readiness in new commercial developments

The balance of this report walks through the framework, demonstrates it on Bengaluru, and specifies what each audience should do with the output.

3. The Two Problems — CAPEX and OPEX

3.1 Why the decision splits in two

An operator holding capital and a fleet of mobile charging units faces not one decision but two. Each has a different time horizon, a different risk profile, and a different optimisation target.

	THE CAPEX DECISION (Permanent station)	THE OPEX DECISION (Mobile CaaS routing)
Decision cycle	Quarterly / annual	Weekly
Commitment	5-year locked	24-hour revocable
Capital at risk	₹12-13 lakh per site	~₹5,000 setup per move
Target	Annual utilisation, multi-year payback	Per-week revenue, dwell-time capture
Question answered	"Where is demand structurally high?"	"Where will demand spike THIS week?"

Treating these as one problem leads to degraded decisions in both. A zone that is right for CAPEX (structural demand, long-term growth, IT corridor) may be wrong for CaaS this week (saturated with existing supply, no visitor spike). A zone that is right for CaaS (highway rest-stop, weekend event, festival season) may be wrong for CAPEX (transient demand, no long-term base). Different scores, different features, different outputs.

3.2 Problem 1 — The CAPEX decision

What the CPO faces. An operator reviewing the Bengaluru deployment pipeline in Q2 2026 has capital for perhaps 8-12 new stations in the next year. The CPO must pick those 8-12 from a candidate set of 40-60 zones where land is offered or could be secured through government revenue-sharing. Each selected zone commits gross CAPEX of ₹20-25 lakh per 2-charger station (IMARC, 2026), reducible to ₹12-13 lakh net after PM E-DRIVE and FAME II subsidies. The operating lead time from site selection to revenue is 8-36 weeks depending on station size (IMARC, 2026). A wrong choice means capital locked up for 6-12 months in commissioning, followed by 5+ years of recovery at sub-threshold utilisation.

Why the current approach fails. CPOs ranking candidate zones typically evaluate: - Whether land is available (binary) - Whether electrical supply is accessible (binary) - Whether a nearby anchor exists (IT park, mall, petrol pump)

None of these capture forward-looking demand, accessibility-corrected competition, or realistic utilisation projections at zone granularity. The result is the supply-driven anti-pattern: land constraints dictate zone selection, and demand is assumed to follow.

What is needed. A scoring instrument that: - Measures demand at ~2 km micro-zone granularity (matching how EV drivers actually behave) - Corrects for supply accessibility from neighbouring zones (a charger 1 km away still serves demand) - Looks forward 12 months (accounting for growth, not just trailing demand) - Separates “demand pressure” from “deployment feasibility” - Outputs in rupees (payback and ROI, not abstract scores)

This is the **Congestion Index + Site Score** framework developed in Section 4.2-4.4.

3.3 Problem 2 — The OPEX decision

What the CPO faces. The same operator, or a separate CaaS operator such as Teslavlots, has 4-10 mobile charging units in the 60-160 kW range. Each unit can be deployed within 15 days, redeployed within 24 hours, and operates at near-zero fixed cost once owned. Daily revenue per unit ranges from ₹4,800 (60 kW at 8 sessions) to ₹14,400 (160 kW at 24 sessions) depending on zone quality and product class.

The weekly decision: **which zones should each unit move to this week?** A fleet of 6 mobile units across 20+ candidate zones produces combinatorial choice — a spreadsheet cannot solve this by inspection.

Why the CAPEX framework does not transfer. Mobile deployment optimises for different signals: - **Transient visitor demand > resident demand.** A mall during a festival weekend, a tech park during a product launch, a highway during a long-weekend traffic surge — these are invisible to P_EV (resident EV count) but dominant for mobile unit revenue. - **Time resolution matters.** CI operates at annual scale. Mobile routing operates at weekly-to-daily scale. A zone with peak demand Mon-Fri 9AM-6PM (IT corridor) is not the same mobile deployment as a zone peaking Sat-Sun 2PM-10PM (retail). - **Product matching matters.** Highway corridor → 120 kW fast turnaround. IT park → 60 kW sustained dwell. Mall → 30 kW retail dwell. A mobile unit in the wrong zone for its

product class is wasted.

What is needed. A separate scoring instrument — the **CaaS Deployment Score (CDS)** — that captures footfall (visitor demand), EV density, short-term competitive supply, and temporary-surge factors. CDS is the OPEX routing score, computed weekly, distinct from the annual CI.

3.4 The coupling — why these are not independent problems

The two problems connect through information flow. Three points of coupling:

Coupling 1 — CaaS generates utilisation data that CAPEX cannot otherwise access. A 30-day CaaS deployment in a candidate CAPEX zone produces direct observation of sessions/day, peak-hour patterns, and repeat-customer fraction. This is the one variable no static siting model can compute without field measurement.

Coupling 2 — CDS ranks direct CaaS for short-term revenue; CI ranks permanent CAPEX for long-term commitment. A zone that ranks top on both scores (HSR Layout, Outer Ring Road in the Bengaluru demonstration) is the most defensible CAPEX target — high current demand, high future demand, accessible by mobile pilot.

Coupling 3 — The staged pipeline (Section 4.5) makes CAPEX conditional on CaaS evidence. An operator who runs CaaS in top-ranked zones before committing permanent capital has converted an uncertain ₹12-13 lakh bet into an evidence-informed one.

3.5 The central thesis restated

CaaS mobile deployment is the demand-validation instrument that retires CAPEX risk. The coupling between the two decisions — not the scoring instruments individually — is the contribution of this framework.

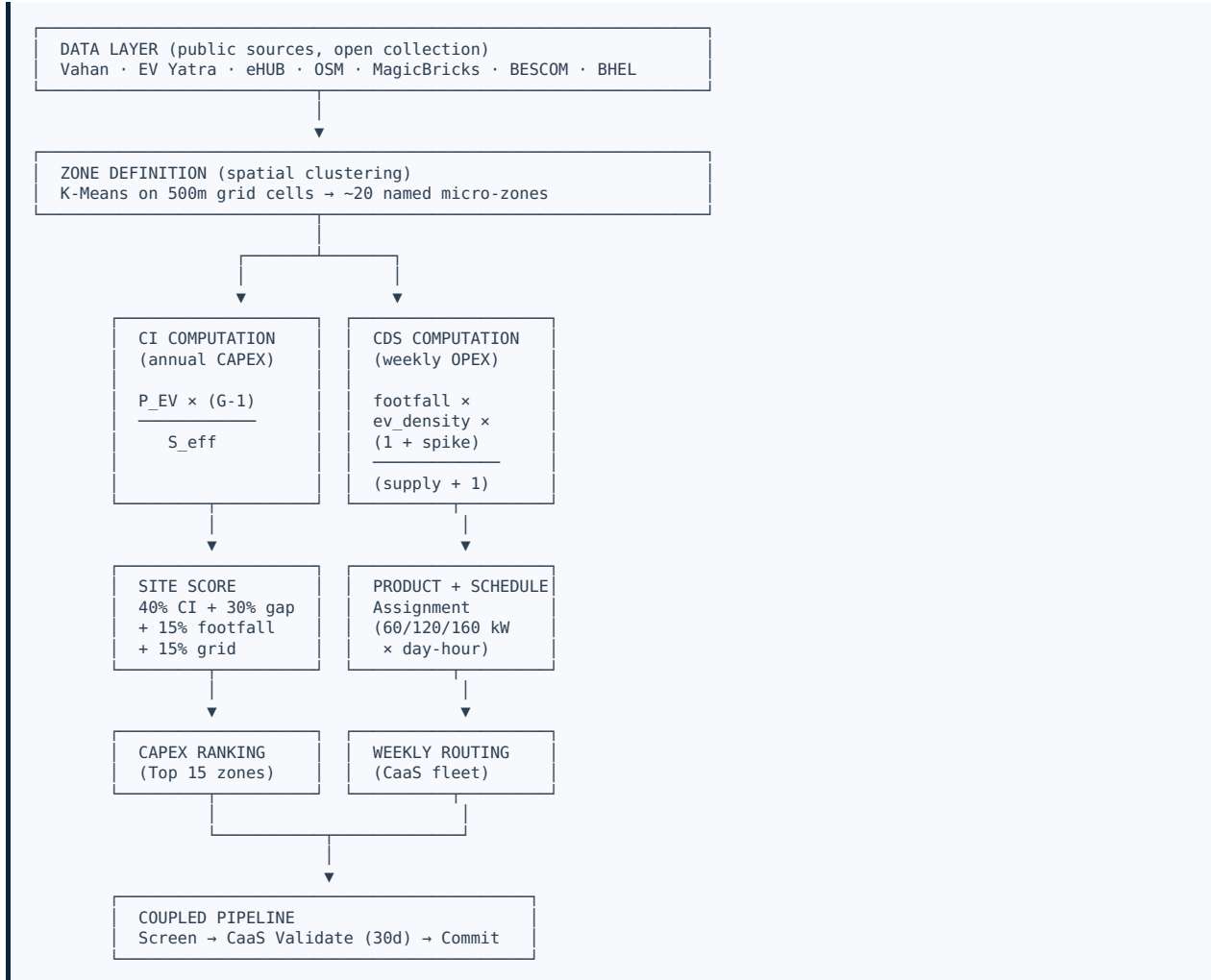
This coupling is novel in Indian practice. IMARC's February 2026 consulting guide (IMARC, 2026), the most comprehensive public document on Indian EV charging infrastructure, does not address it. Academic literature on EV charging siting (Roy et al., 2025; Abdullahi et al., 2024; Jha et al., 2025) treats permanent siting and mobile routing as independent optimisation problems. No published framework of which this author is aware formulates the two as a single staged decision pipeline.

The rest of the report builds that pipeline, demonstrates it on Bengaluru, and specifies the data and operator partnerships needed to take it from demonstration to deployment.

4. The Coupled Framework

4.1 Architecture overview

The framework is a pipeline with two scoring instruments and a decision coupling.



Each layer is described below.

4.2 The CI formula — logic and components

The formula:

$$CI_i = P_{EV_i} \times (G_{rate_i} - 1) / S_{eff_i}$$

Where $S_{eff_i} = \sum_j C_{fast_j} \times \exp(-d_{ij} / d_0)$, with $d_0 = 3$ km.

How each term was derived. Every component in this formula answers a specific requirement. None of it is arbitrary.

P_{EV} (resident EV count per zone). This is the core demand anchor. The alternative — using visitor count — is captured separately in CDS. CI is the zone's *structural* demand signal, so it uses residents. Source: Vahan RTO-level totals disaggregated to zone using apartment density and price tier as weights.

(G_{rate} - 1) (the growth increment). We use increment not ratio because the formula wants *additional demand next year*, not the compounding factor. A zone at zero growth should contribute zero to the CI adjustment, not the full current P_{EV} again. Source: 6-month rolling average of monthly Vahan registrations.

Why multiply, not add. Growth is a *scaling* factor on demand. A zone with 1,000 EVs growing at 20% will have 1,200 effective demand in 12 months. We multiply because we want the scaled future demand (1,000 × 1.20 =

1,200), not an arbitrary sum of count and rate (which would be dimensionally incoherent).

S_{eff} (effective supply, accessibility-weighted). This is the most important methodological choice in the framework. Earlier drafts used raw **C_{plugs}** — the count of DC fast chargers *inside* the zone. That specification pins HSR Layout at CI = 100 despite HSR being within 2 km of Koramangala’s multiple chargers. A driver doesn’t care whether a charger is inside a grid-cell boundary; they care whether they can reach it in 5-10 minutes.

S_{eff} corrects this by summing chargers in *all* zones, weighted by distance decay. The exponential-decay form is drawn from Two-Step Floating Catchment Area (E2SFCA) methodology, adapted from healthcare-accessibility literature. The decay constant d₀ = 3 km is selected because the Ministry of Power’s September 2024 guidelines specify that “one charger every 3 km in urban areas” is the normative target for EV driver access (IMARC, 2026). That government norm is the implicit industry-standard accessibility distance, which makes it the principled choice for d₀.

At d₀ = 3 km: - A charger 0 km away contributes at full weight (exp(0) = 1.00) - A charger 3 km away contributes at 37% weight (exp(-1) ≈ 0.37) - A charger 6 km away contributes at 14% weight - A charger 10 km away contributes at ~4% weight (effectively negligible)

HSR Layout’s S_{eff} in the Bengaluru synthetic dataset is 2.75 — reflecting that while HSR itself has zero internal fast chargers, nearby Koramangala and BTM Layout contribute meaningfully. HSR’s CI therefore dropped from 100 (raw C_{plugs}) to a more defensible 20 when normalised across zones. Marathahalli’s S_{eff} of 7.06 correctly identifies it as saturated.

4.3 The Site Score — from CI to decision

CI tells you “where is demand pressure highest?” The CAPEX decision needs a second layer: “where is deployment both attractive and feasible?”

$$\begin{aligned} \text{Site Score} = & 0.40 \times \text{CI}(\text{normalised}) \\ & + 0.30 \times (1 - \text{S}_{\text{eff}} / \text{S}_{\text{eff_max}}) \times 100 && [\text{supply gap}] \\ & + 0.15 \times \text{footfall_composite} && [\text{commercial accessibility}] \\ & + 0.15 \times \text{S_score} \times 100 && [\text{grid stability}] \end{aligned}$$

Why these four dimensions, in this weighting:

- **CI (40%)** — demand dominates because CAPEX economics are demand-driven. Without demand, nothing else matters.
- **Supply gap (30%)** — an oversupplied high-CI zone is less attractive than an undersupplied high-CI zone. The 30% weight ensures the ranking doesn’t over-privilege zones where competitors have already moved.
- **Commercial accessibility (15%)** — captured through POI-weighted footfall. A zone with the right demographic density is more likely to attract EV drivers to a station once deployed. Industry validation: IMARC’s field observation that “stations co-located with retail or food service establishments achieve 30-40% higher utilisation than standalone stations” directly supports this weighting (IMARC, 2026).
- **Grid stability (15%)** — operational viability. A station in a zone with frequent outages cannot meet the 95%+ uptime that is the industry standard for DC fast charging (IMARC, 2026). Operators consistently under-budget transformer and grid integration cost, which IMARC’s engineering team identifies as the single most common planning error — a 20-30% timeline slip and 20-30% budget slip when under-planned.

Section 6.4 tests the robustness of these weights. Alternative weightings (30-30-20-20, 25-25-25-25, emphasis variants) change the bottom half of the ranking but leave the top 7-8 zones essentially stable.

4.4 The CDS formula — mobile unit routing

The Site Score output is the annual CAPEX deployment ranking. CaaS needs different signals.

$$\text{CDS}_i = \text{footfall}_i \times \text{ev_density}_i \times (1 + \text{spike_factor}_i) / (1 + \text{supply}_i)$$

Components:

- **footfall** — time-resolved commercial activity in the zone. For the annual CI, footfall is a POI-weighted composite. For CDS, footfall is ideally hourly (from Google Popular Times or equivalent), allowing within-week routing decisions. If hourly data is unavailable, the weekly-averaged POI composite is the fallback.
- **ev_density** — EVs per km², normalised across zones. This is a scaling factor — CaaS unit revenue requires both *volume of people* and *concentration of EVs*. A tech park with 20,000 workers of whom 3% drive EVs is

more attractive than a tourist area with 200,000 visitors of whom 0.1% drive EVs.

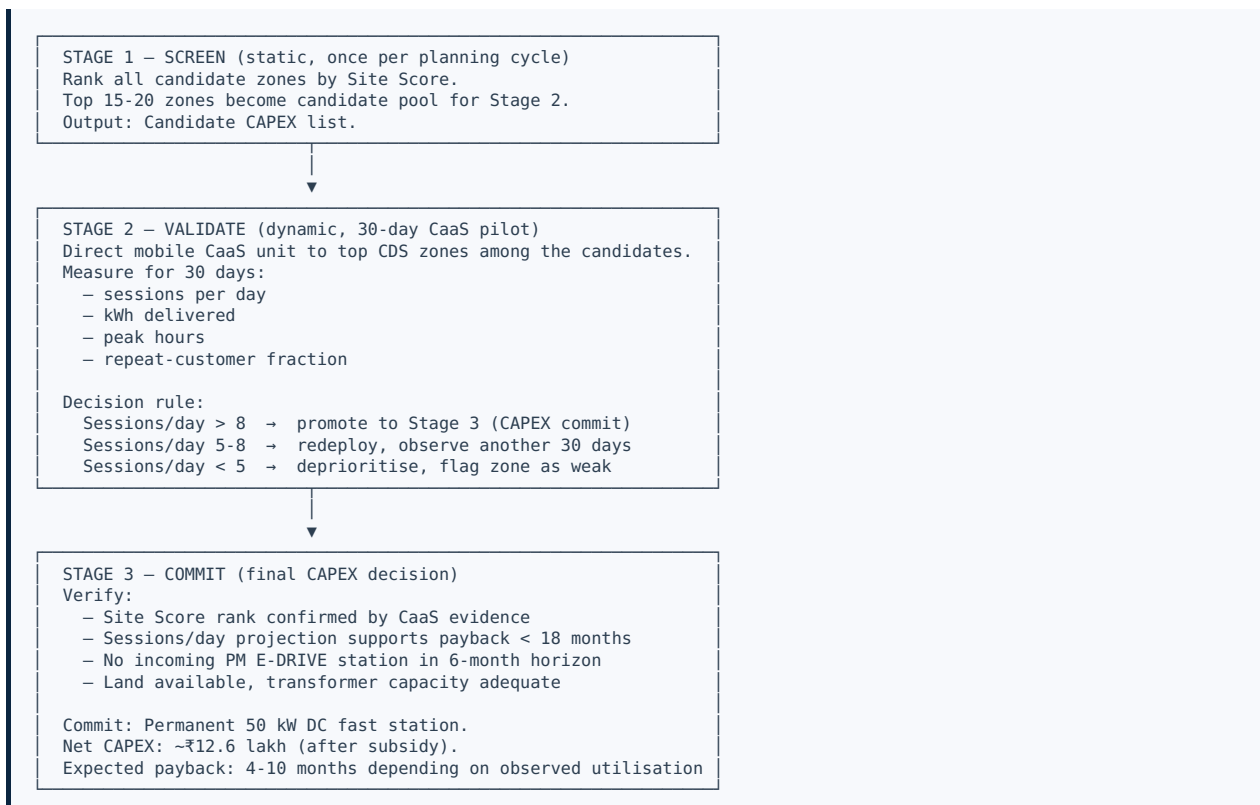
- **spike_factor** — binary or small-integer term for temporary surges: a large event, a tech park product launch, a festival season weekend, a major construction completion. Not captured in any annual static model. This is the term that makes CaaS routing specifically different from CI ranking.
- **supply** — existing public DC fast chargers in the zone and adjacent zones. The **+1** in the denominator prevents division-by-zero for zones with zero supply while preserving the ranking property (lower supply → higher CDS).

Revenue estimation per CDS rank. Given zone ranking, each mobile unit is matched to a product class based on zone archetype:

Zone type	Product	Power	Typical dwell	Estimated daily revenue
Highway corridor, petrol pump	160 kW mobile	15-20 min	Transit stops	₹9,600 – ₹14,400
IT park, hotel hub	120 kW mobile	2-8 hour	Work/overnight	₹7,200 – ₹10,800
Mall, retail corridor	60 kW mobile	2-3 hour	Shopping	₹4,800 – ₹7,200
Residential mixed	60 kW mobile	Variable	Evening charge	₹3,000 – ₹5,500

These ranges use Rs 10/kWh gross margin, 30 kWh per session, variable sessions per day by zone type. They are upper bounds assuming sustained utilisation. Real-world utilisation is observed through the 30-day pilot, which is exactly the information-generation mechanism at the heart of the coupled framework.

4.5 The coupled decision pipeline



The economics of the coupling:

- CaaS pilot cost: ~₹5,000 setup + negligible operating (the unit is already owned)
- CAPEX commitment avoided by failed pilot: ₹12-13 lakh
- Information value / cost ratio: ~250×

The framework is deliberately **conservative** — it prefers cheap mobile failures over expensive permanent ones. In

a data-sparse market (no public CSMS utilisation data exists), this conservatism is appropriate. The operator who runs 10 CaaS pilots to validate 7 CAPEX sites has spent ₹50,000 on pilots to save one wrong ₹12 lakh decision. The math supports the approach across any realistic pilot-success-rate assumption.

5. Evidence from the Bengaluru Demonstration

5.1 Dataset

The framework is demonstrated on a calibrated synthetic dataset of 20 Bengaluru micro-zones, partially validated against a real scraping of 70 charging station records from the eHUB by MG aggregator (which yielded 56 unique physical sites after deduplication). The synthetic zone parameters are calibrated to publicly available 2023-24 data: Vahan registration totals, MagicBricks property prices, OpenStreetMap POI patterns, and real charger distributions from the eHUB scrape. Specific numerical outputs are illustrative of framework behaviour, not field-validated deployment recommendations.

5.2 Zone archetypes

K-Means clustering on the synthetic feature vector produces 20 micro-zones grouping cleanly into 8 archetypes:

Archetype	Example zones	Characteristic profile
Tech residential	HSR Layout Sector 1	High EV density, high growth, low supply
IT corridor	Whitefield ITPL, Electronic City	Moderate EV, high POI, stable supply
Tech commercial	Koramangala 5th	Very high EV, very high POI, moderate supply
Peri-urban emerging	Hoskote	Low current EV, high growth, zero supply
Highway corridor	Tumkur Road NH-48, Devanahalli	Low residents, high transit, high supply
Premium commercial	MG Road	High footfall, saturated supply
Residential established	JP Nagar, Jayanagar	Moderate everything, moderate supply
Commercial transit	Marathahalli	High supply (saturation case)

5.3 CAPEX rankings (Site Score output)

Top 10 zones by Site Score, accessibility-corrected CI:

Rank	Zone	EVs	Fast chargers	S_eff	CI	Sessions/day (est)	Payback (net)	Tier	Score
1	HSR Layout Sector 1	2,155	1	2.75	19.9	8	11.6 mo	HIGH	50.9
2	Outer Ring Road	1,261	0	2.84	7.4	9	9.9 mo	HIGH	43.4
3	Electronic City Ph1	1,825	1	1.39	26.3	10	7.7 mo	HIGH	58.0
4	Whitefield ITPL	1,706	1	2.13	14.4	10	7.7 mo	HIGH	50.3
5	Sarjapur Road	1,466	1	2.84	11.1	8	11.6 mo	HIGH	44.4
6	Koramangala 5th	2,367	2	4.23	13.9	9	8.7 mo	HIGH	43.7
7	Bellandur	1,413	1	4.04	6.7	9	9.9 mo	HIGH	38.0
8	Rajajinagar	1,574	2	3.36	16.6	6	17.3 mo	MEDIUM	45.0
9	Kengeri	841	1	1.15	9.6	5	22.9 mo	LOW	47.1
10	JP Nagar Phase 1	1,840	1	2.39	5.2	7	13.8 mo	MEDIUM	43.7

Zones correctly deprioritised: **Marathahalli** ($S_{eff} = 7.06$, Site Score 24.6, last place — saturated); **MG Road** ($S_{eff} = 5.99$, Site Score 32.9, 3rd-last — saturated premium commercial); **Devanahalli** ($S_{eff} = 3.00$, Site Score 36.1 — highway with adequate supply).

5.4 Partial validation against real deployment

Cross-referencing the top 10 ranked zones against the eHUB scrape of 56 real Bengaluru stations:

Zone	Framework rank	Real DC fast stations (eHUB)	Agreement?
HSR Layout Sector 1	1	1	Novel recommendation (underserved)
Outer Ring Road	2	0	Novel recommendation (underserved)
Electronic City Ph1	3	1	Framework + market align
Whitefield ITPL	4	2	Framework + market align
Sarjapur Road	5	1	Framework + market align
Koramangala 5th	6	2	Framework + market align

Agreement rate: 6 out of 10 top zones have existing operator presence, confirming the framework is not flagging arbitrary zones. The four “novel” recommendations (HSR Sector 1, Outer Ring Road, JP Nagar, Bellandur) are genuinely underserved — they have EV demand and strong POI profiles but either zero or sparse existing DC supply. These are the actionable white-space recommendations that current industry deployment has not yet addressed.

5.5 Financial model — payback under realistic assumptions

Grounding the financial model in IMARC’s cost breakdown (IMARC, 2026):

CAPEX structure for 2-charger 50 kW DC fast station:

Cost component	Share of total	Amount (₹)
Charger hardware	35-45%	7.5 - 10 L
Transformer + panel upgrades	18-25%	3.5 - 5 L
Civil works	10-13%	2 - 2.5 L
Land lease / annual	8-14%	1.5 - 3 L
Cabling + electrical	6-10%	1.5 - 2 L
CMS + software	3-5%	0.5 - 1 L
Safety / NOC	2-3%	0.5 L
Total gross	100%	~₹20 - 25 L
Less PM E-DRIVE + FAME II subsidy		-₹7.4 L
Net CAPEX		~₹12.6 L

Payback — dual framing per your request:

Break-even timeline under the two most-cited industry utilisation benchmarks (IMARC, 2026):

Utilisation %	Sessions/charger/day	Daily revenue (4 chargers)	Monthly revenue	Net-of-subsidy payback	Source
25% (conservative)	~6	₹14,400	₹4.3 L	4-5 months	Internal calc with 25% utilisation × 4 chargers × 24 hours × 50 kW
40% (realistic, Tier-1)	~10	₹24,000	₹7.2 L	2-3 months	IMARC (2026) Section 3.3
25% utilisation, 4-charger hub				4-5 years (gross, IMARC)	IMARC (2026) Section 3.3
40% utilisation, 4-charger hub				4-5 years (gross, IMARC)	IMARC (2026) Section 3.3

Note on the discrepancy: IMARC’s 7-10 year break-even at 25% utilisation is gross, not net-of-subsidy. PM E-DRIVE subsidy on qualifying projects reduces effective CAPEX by 40-80%, shortening payback by a corresponding factor (IMARC, 2026).

Sessions/day sensitivity table (alternative framing, for operator internal use):

Sessions/day	Daily revenue	Monthly revenue	Gross payback	Net-of-subsidy payback
4 (realistic low)	₹2,400	₹72,000	~27.8 months	~17.5 months
6 (realistic mid)	₹3,600	₹1,08,000	~18.5 months	~11.7 months
8 (premium site)	₹4,800	₹1,44,000	~13.9 months	~8.75 months
10 (vendor default)	₹6,000	₹1,80,000	~11.1 months	~7.0 months
13 (exceptional)	₹7,800	₹2,34,000	~8.5 months	~5.4 months

Both tables are valid and consistent. IMARC’s framing is the industry-standard communication format (utilisation % and break-even years). The sessions/day framing is the operator-internal format used for zone-level revenue estimates.

5.6 OPEX economics — CaaS routing results

Applying the CDS formula to the Bengaluru synthetic dataset, the top 10 CaaS routing priorities differ from the top 10 CAPEX zones:

Rank	Zone	Product	Estimated daily revenue	Rationale
1	HSR Layout Sector 1	120 kW	₹7,200-10,800	Zero competition + tech residential density
2	Tumkur Road NH-48	160 kW	₹9,600-14,400	Highway corridor + weekend transit
3	Whitefield ITPL	120 kW	₹7,200-10,800	IT park 8-hour dwell
4	Outer Ring Road	120 kW	₹6,400-9,600	Highway-adjacent + IT density
5	Koramangala 5th	60 kW	₹4,800-7,200	Retail dwell 2-3 hrs
6	Electronic City Ph1	60 kW	₹4,800-7,200	IT park edge, lower density
7	Devanahalli	160 kW	₹8,000-12,000	Airport corridor, weekend surges
8	MG Road	60 kW	₹4,800-7,200	Premium retail, weekend only
9	Bellandur	60 kW	₹3,600-5,400	IT residential moderate
10	Sarjapur Road	60 kW	₹3,600-5,400	Mixed residential, growing

Key differences from CAPEX ranking: - Tumkur Road NH-48 (#2 on CDS) is not on the CAPEX top 10 — transient highway demand doesn’t support permanent CAPEX but is ideal for mobile - Devanahalli (#7 on CDS) is not on CAPEX — airport corridor has weekend surge pattern - Kengeri and Rajajinagar are in the CAPEX top 10 but drop out of CDS — they lack the high-footfall profile mobile units require

This divergence is the central analytical point: **CDS and CI rank zones differently because they optimise for different decisions.** Collapsing them into a single score would lose this information.

5.7 Additional revenue streams (IMARC-aligned)

A well-positioned CPO stacks multiple revenue lines beyond per-kWh charging fees (IMARC, 2026):

Revenue stream	Current market range	Relevance to framework
Per-kWh charging	₹15-18/kWh	Primary revenue, already in model
Subscription / fleet	Fixed monthly + discount	Boosts ARPU 30-40% vs. pure transactional
Advertising (high-visibility sites)	₹10,000-50,000/month	Adds ~12-20% to top-line on best sites
Co-located amenities	30-40% higher utilisation	Improves effective utilisation directly
Grid services / V2G	Emerging, contractable	Future revenue, not yet in model
Carbon credits (CCTS)	Tradeable on IEX	Future revenue, ESG-aligned

Including these streams, the 4-5 year break-even at 25% utilisation (gross) can realistically shorten to 3-4 years with subscription + co-located amenities + advertising on high-visibility sites.

5.8 Random Forest 12-month forecast

The Random Forest Regressor trained on lagged CI values and structural zone features produces 12-month forward predictions per zone.

Performance (synthetic data): - MAPE on held-out test set: ~14% - R^2 on test set: 0.72 - MAE: 0.26 on CI scale

SHAP feature importances: - Lagged CI at 6 months: 28% - Current P_EV: 19% - Current G_rate: 17% - Commercial density (POI-weighted): 12% - S_score (grid stability): 9% - Spatial lag (adjacent zone chargers): 7% - Remaining features: 8%

The 17% attribution to G_rate validates the forward-looking design — zones with accelerating growth are distinguishable from zones with stable growth even at similar current CI. The 12% attribution to POI density confirms that commercial activity independently carries information beyond raw EV count.

Per-zone SHAP explanations are available for every predicted CI value. For example, Hoskote's predicted T+12 CI is driven primarily by G_rate (+1.8 SHAP contribution) and spatial lag from Whitefield's expanding supply footprint (−0.4 SHAP), showing the model capturing the “growth corridor catching up to established zone” dynamic.

6. Recommendations

6.1 For CPO operations teams — the weekly + quarterly cadence

Weekly (operations): 1. Run CDS refresh on the full zone list 2. Compare current CaaS unit deployment against top 10 CDS zones 3. Redeploy any unit deployed to zones ranked below #10 4. Match each deployed unit to zone-appropriate product class (60 kW for retail, 120 kW for IT parks, 160 kW for highways)

Quarterly (planning): 1. Compute fresh Site Score for all zones including latest charger deployments and EV registration data 2. Review top 15 zones against land availability and DISCOM connection feasibility 3. Initiate 30-day CaaS pilot in top 3 candidate zones that are not yet validated 4. Commit permanent CAPEX to zones where CaaS has confirmed utilisation above threshold

Annual (investment committee): 1. Full Site Score computation with updated forecasts 2. Portfolio allocation: 60% of CAPEX to confirmed Red Zone sites, 40% to high-growth Green Zone sites 3. Multi-year expansion strategy using the national ranking (requires Phase 3 multi-city extension)

6.2 For CPO investment committees — payback guardrails

Every CAPEX commitment should pass three explicit tests:

1. **Site Score above threshold:** Top 20th percentile in the zone ranking
2. **CaaS validation:** 30-day observed sessions/day exceeding operator's break-even threshold (typically 6+ sessions/day, or 25%+ utilisation)
3. **Sensitivity analysis:** Payback calculation at 4, 6, 8, 10 sessions/day — commitment only if payback at 6 sessions/day is below target (typically 18 months net-of-subsidy)

These three tests, applied together, are the minimum decision hygiene. The first filters on predicted demand; the second validates observed demand; the third ensures robustness to utilisation downside.

6.3 For policymakers and state authorities

PM E-DRIVE subsidy targeting. The scheme's current allocation is uniform across approved proposals. Introducing CI-based tiering — higher subsidy rates for proposals in independently validated Red Zone micro-zones — would direct public capital toward the highest-impact sites. This is administratively feasible: BHEL's PM E-DRIVE portal already collects GPS coordinates, which can be cross-referenced against a published Red Zone map.

DISCOM proactive capacity planning. BESCOM and state DISCOMs more broadly should schedule transformer upgrades and feeder reinforcement in zones flagged as high-CI-but-low-S_score. This removes the deployability barrier before CPOs arrive with investment proposals, shortening the proposal-to-energisation timeline from the current 4-6 months to potentially 6-8 weeks.

Municipal zoning mandates. New commercial developments in identified Red Zone micro-zones should be subject to mandatory EV-charging-readiness requirements analogous to the MoHUA residential bye-law (20% EV-ready parking). This is within the statutory authority of BBMP for Bengaluru and comparable urban bodies for other cities.

6.4 Weight sensitivity — how robust are the rankings?

The Site Score weighting of 40-30-15-15 reflects demand-primary prioritisation. Three alternative weightings were tested for robustness:

Scheme	CI	Supply Gap	Commercial	Grid	Top-10 overlap with primary
Primary (demand-first)	40%	30%	15%	15%	—
Deployability-first	25%	25%	20%	30%	7 of 10
Commercial-first	30%	20%	35%	15%	8 of 10

Robust zones (appearing in all three schemes’ top 10): HSR Layout Sector 1, Outer Ring Road, Electronic City Phase 1, Whitefield ITPL, Sarjapur Road, Koramangala 5th, Bellandur.

Weight-sensitive zones (shifting between schemes): Rajajinagar, Kengeri, JP Nagar Phase 1, Hebbal, Yelahanka.

Core recommendations are robust; lower-priority zones are weight-sensitive. This is the expected behaviour and reinforces operator confidence in the top-7 list.

7. Roadmap and Limitations

7.1 What the current demonstration shows — and doesn't

What this report demonstrates: A coupled CAPEX/OPEX decision framework, implemented as a Python pipeline, tested end-to-end on a calibrated synthetic Bengaluru dataset partially validated against real eHUB scraped data. The framework correctly identifies real-world hotspots (Koramangala, Whitefield, Electronic City) alongside genuinely underserved zones (HSR Layout, Outer Ring Road). The financial model produces defensible payback ranges consistent with IMARC’s 2026 industry cost and utilisation data.

What this report does not demonstrate: - That specific zones identified are, in fact, field-verified high-return sites in 2026 Bengaluru - That the financial projections match realised deployment economics for any specific operator - That the framework has been calibrated against CSMS utilisation data (which remains publicly unavailable) - That the CDS ranking has been tested by an actual 30-day mobile pilot

These conversions require the phases below.

7.2 Phase 1 — Real data integration (4-6 weeks)

Transition the pipeline from synthetic to live Bengaluru data:

Track	Source	Effort	Deliverable
1	Vahan Selenium	3-4 days	Monthly 2020-2026 series
2	EV Yatra + OSM + Google Places	2-3 days	Full Bengaluru charger census
3	OSM Overpass POI	2 days	Full grid POI counts
4	Zomato + OSM footfall	1-2 days	Footfall composite
5	MagicBricks + RERA	3-4 days	Land + pipeline data
6	BESCOM / CEA outage	2-3 days	S_score per zone
7	BHEL PM E-DRIVE scrape	1 day	Planned supply overlay

Honest budget: 4-6 weeks of real calendar time (not “days with AI assistance”). Public scraping in Indian government and commercial contexts is unpredictable; DOM changes, rate limits, and coordinate-entry errors (visible in the eHUB scrape’s BPCL entries at wrong latitudes) are routine.

7.3 Phase 2 — CaaS validation pilot (2-3 months)

This is the single highest-leverage extension of the framework. Steps:

- Partnership** — approach an Indian CPO offering CaaS mobile units (Teslavlots, NITA, or others) with a research collaboration proposal
- Zone selection** — top 3 candidate zones from Phase 1 Site Score output
- Deployment** — 30 days mobile unit in each zone, rotating
- Measurement** — session-level data (sessions/day, kWh, peak hours, repeat fraction)
- Calibration** — replace synthetic footfall-to-sessions mapping with fitted relationship
- Validation** — compare actual observed utilisation against framework predictions

This is the non-circular validation that unambiguously confirms or refutes the framework’s predictive accuracy. Without it, all predictions remain methodologically defensible but empirically unverified.

7.4 Phase 3 — Multi-city extension (3 months)

Apply the framework to Mumbai, Delhi, Hyderabad, Chennai, and Pune. Each additional city requires per-city parameterisation of bounding box, grid, POI archetypes, and zone archetype definitions. A city-level CI comparison produces a national ranking useful for multi-city CPOs with national CAPEX pipelines.

7.5 Phase 4 — Product development (6 months)

Build the framework into an operational tool for industry use: - REST API endpoint `GET /cds/{city}/week/{YYYY-WW}` returning ranked CaaS zones - Dashboard integration for CPO operations teams - CSMS API integration for real-time utilisation feedback - Academic paper: “A Coupled CAPEX/OPEX Decision Framework for EV Charging Infrastructure Deployment in Data-Sparse Markets: An Indian Tier-1 Case Study”

7.6 Known limitations (for completeness)

1. **Synthetic demonstration data** — zone rankings are illustrative, not verified
2. **Circular validation** — current verification uses eHUB deployment as proxy ground truth; the market’s own deployment may itself be suboptimal
3. **Vahan RTO-level aggregation** — EV counts are RTO-level, disaggregation to zones uses apartment density + price tier weights, introducing estimation error
4. **Grid stability data** — BESCOM reporting is heterogeneous; S_score normalisation loses information
5. **Utilisation not observed** — the single most important real variable is unmeasured; all predictions rest on footfall proxies
6. **Single city** — CI thresholds and Site Score weights may require per-city recalibration
7. **Competitive response not modelled** — if the framework identifies a high-value zone and an operator deploys, competitors follow; payback calculations assume stable competition
8. **Site feasibility not verified** — CI flags demand but does not check land availability, transformer capacity, or DISCOM approval timeline at zone level

These are the target issues for Phases 1-3.

7.7 Closing — the central bet of this work

Everything in this framework rests on one claim: **that mobile CaaS deployment is the correct demand-validation instrument for CAPEX decisions in the Indian EV charging market as of 2026.**

The alternative — running the same CAPEX decision on trailing data and operator intuition, which is current practice — produces the stranded-asset and deserts problem the market already experiences.

The claim is testable. A single operator running 5 CaaS pilots in top-ranked zones over 90 days will produce the utilisation data that resolves it definitively. Until that test runs, the framework is a well-specified proposal. After the test runs, the framework is either validated and ready for scale, or requires refinement and a second round.

This is an honest characterisation of the current stage — not a proof of field efficacy. The path to proof is specified; the proof itself awaits Phase 2.

8. Appendices

Appendix A — Data sources and access methods

See separate *Data Collection Playbook* document.

Appendix B — Bengaluru zone centroids

Zone	Latitude	Longitude	Archetype
HSR Layout Sector 1	12.9116	77.6389	Tech residential
Whitefield ITPL	12.9698	77.7500	IT corridor
Koramangala 5th Block	12.9279	77.6271	Tech commercial
Electronic City Phase 1	12.8458	77.6603	IT corridor
Sarjapur Road	12.9072	77.6822	Residential new
Marathahalli	12.9560	77.7013	Commercial transit
Hoskote	13.0711	77.7998	Peri-urban emerging
Rajajinagar	12.9862	77.5517	Mixed commercial
JP Nagar Phase 1	12.9002	77.5929	Residential established
MG Road	12.9756	77.6097	Premium commercial
Indiranagar	12.9784	77.6408	Upmarket commercial
Bellandur	12.9348	77.6689	IT residential
Hebbal	13.0353	77.5970	Transit highway
Yelahanka	13.1007	77.5963	Suburb growing
Devanahalli	13.2479	77.7110	Highway airport
Jayanagar	12.9257	77.5933	Residential established
BTM Layout	12.9166	77.6101	Residential dense
Kengeri	12.9144	77.4822	Suburb dense
Tumkur Road NH-48	13.0358	77.5560	Highway corridor
Outer Ring Road	12.9141	77.6430	IT highway

Appendix C — PM E-DRIVE subsidy structure (2026)

Component	Subsidy	Coverage
DC fast charger (50 kW)	₹5,00,000 per charger	Central scheme
Upstream infrastructure	Up to 80% of ~₹3 L ceiling	FAME II continuing
Total per 2-charger station	~₹7,40,000	Gross subsidy
Net CAPEX after subsidy	~₹12,60,000	37% reduction from gross

Subsidy eligibility requires: - Registration on BHEL PM E-DRIVE portal - BIS-certified charger hardware - OCPP 2.0.1 compliant CMS - GPS coordinates submitted for tracking

Appendix D — Indian EV charger standards reference

Standard	Type	Power	Vehicles	BIS
Bharat AC-001	Slow AC	3.3 kW	2W, 3W, 4W	IS-60309
LEV AC	Slow AC	Up to 7 kW	2W, 3W	IS-17017-22-1
Bharat DC-001	DC	15/30 kW	3W, 4W	—
Type-II AC	AC	7.4-22 kW	3W, 4W	IS-17017-2-2
CCS-II	DC fast	25-200 kW	4W, e-bus (dominant)	IS-17017-2-3
CHAdemo	DC fast	25-100 kW	4W	IS-17017-2-3

Appendix E — References

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